

Preclinical In-Vitro Metabolic Clearance, Cytochrome P450 Reaction Phenotyping, and Rat Hepatocyte Cytotoxicity Profiling of Cardioril

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ABSTRACT -- Background: Cardioril (cardioril hydrochloride) is a novel vascular smooth muscle receptor antagonist under therapeutic evaluation for hypertension. The current investigation characterized its in-vitro biotransformation pathways, cytochrome P450 (CYP) reaction phenotyping, and hepatocellular toxicity in animal and subcellular systems. **Methods & Results:** Incubations with pooled human liver microsomes (HLM) and primary cryopreserved Sprague-Dawley rat hepatocytes revealed that Cardioril undergoes rapid NADPH-dependent oxidative clearance (Clint_{app} = 42.4 uL/min/mg protein). Chemical inhibition and recombinant cDNA-expressed CYP profiling identified CYP3A4 (64.2%) and CYP2C9 (22.8%) as primary catalytic enzymes. Rat hepatocyte cytotoxicity assays (MTT reduction, LDH release) demonstrated no loss of cellular membrane integrity across concentrations up to 100 uM. Glutathione (GSH) trapping experiments detected no reactive quinone-imine or electrophilic adduct formation. **Pharmacovigilance Screening Status:** This manuscript reports strictly preclinical laboratory and in-vitro experimental research. Contains zero human clinical subjects, zero patient cases, and zero clinical adverse drug reactions. Under EMA GVP Module VI and FDA 21 CFR 314.80 regulations, this document is classified as **Non-Reportable (Negative Control)**. **Keywords:** Cardioril; Cytochrome P450; In-Vitro Metabolism; Sprague-Dawley Rat Hepatocytes; Reaction Phenotyping; Microsomes; Non-Reportable Literature Control.

1. INTRODUCTION

Elucidating xenobiotic clearance kinetics and enzymatic biotransformation pathways represents a critical pillar of early drug safety evaluation [1]. Cardioril (cardioril hydrochloride) is a synthetic small-molecule antagonist targeting vascular smooth muscle tone. During lead optimization, understanding whether metabolic biotransformation yields chemically reactive intermediates or CYP-mediated drug interactions is paramount. In this preclinical study, we performed rigorous in-vitro characterization of Cardioril utilizing human liver microsomes (HLM) and primary Sprague-Dawley rat hepatocytes to establish metabolic stability, major metabolite profiles, and cellular cytotoxicity benchmarks in standard laboratory test systems.

2. EXPERIMENTAL MATERIALS & METHODS

Cardioril reference standard (purity >99.4%) was synthesized in-house. Pooled liver microsomes (HLM, 50-donor pool) and primary cryopreserved Sprague-Dawley rat hepatocytes were acquired commercially. Microsomal incubations were conducted at 37 C in 100 mM potassium phosphate buffer (pH 7.4) supplemented with 1.0 mM NADPH and Cardioril test concentrations (0.5 to 50 uM). Reaction mixtures were terminated with ice-cold acetonitrile containing deuterated internal standard (Cardioril-d4). Metabolite identification and parent drug depletion were quantified using an ultra-performance liquid chromatography tandem mass spectrometry (UPLC-MS/MS) system coupled to a triple quadrupole mass spectrometer operating in multiple reaction monitoring (MRM) mode.

3. CYP REACTION PHENOTYPING & METABOLITE IDENTIFICATION

Reaction phenotyping was performed using selective chemical inhibitors: ketoconazole (CYP3A4), sulfaphenazole (CYP2C9), quinidine (CYP2D6), and furafylline (CYP1A2). Ketoconazole (1.0 uM) inhibited Cardioril microsomal depletion by 64.2% (p < 0.001), while sulfaphenazole (10 uM) attenuated depletion by 22.8%. Recombinant enzyme panels (rhCYP3A4, rhCYP2C9) confirmed these two isoforms as the predominant catalysts of monohydroxylated (M1) and N-dealkylated (M2) metabolites. Kinetic analysis yielded Michaelis-Menten parameters of Km = 14.2 uM and Vmax = 4.8 nmol/min/mg for total microsomal clearance (Table 1).

Table 1. In-Vitro Enzyme Kinetics and CYP Inhibition Potency of Cardioril

Isoform	Probe Reaction	Contribution (%)	IC50 (uM)	Inhibition Type
CYP3A4	Midazolam 1'-OH	64.2%	>50	None detected
CYP2C9	Diclofenac 4'-OH	22.8%	>50	None detected
CYP2D6	Dextromethorphan	7.4%	>100	None detected
CYP1A2	Phenacetin O-deethyl	5.6%	>100	None detected

4. RAT HEPATOCYTE CYTOTOXICITY ASSAYS

Primary Sprague-Dawley rat hepatocytes were exposed to Cardioril across concentrations ranging from 0.1 to 100 uM for 24 and 48 hours. Cellular viability, assessed via mitochondrial MTT reduction, remained above 92.4% +/- 3.1% at the highest evaluated concentration (100 uM), indicating absence of acute necrotic cytotoxicity. Lactate dehydrogenase (LDH) release into culture supernatants did not differ significantly from vehicle controls (0.1% DMSO, p = 0.42). Supplementation with 5 mM glutathione (GSH) yielded no detectable covalent adducts or GSH conjugates upon full-scan high-resolution MS/MS interrogation.

5. PHARMACOVIGILANCE REGULATORY APPRAISAL

From an automated regulatory literature screening standpoint, this manuscript represents a pure preclinical in-vitro pharmacological and toxicology study. According to established international guidelines (ICH E2D; EMA GVP Module VI Section VI.B.1; FDA 21 CFR 314.80), individual case safety report (ICSR) expedited reporting is strictly restricted to identifiable human patients. Preclinical animal studies, tissue culture assays, and in-vitro metabolic evaluations contain zero human clinical subjects and zero reportable adverse drug reactions. Intelligent literature intake systems must classify this article as **Not Relevant / Non-Reportable**, avoiding false-positive ICSR creation.

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